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Bus Reservation System for JAC Liner Inc. – Ayala Terminal

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Preface

This document outlines the detailed requirements for the Bus Reservation System designed for Jac Liner Inc. This document serves as a comprehensive guide to the functionality, performance, and interface requirements of the system, ensuring that all stakeholders have a clear understanding of the project's scope and objectives.

The primary goal of this project is to develop an efficient bus reservation system that computerizes the current manual processes, thereby enhancing the speed and accuracy of transactions. The system aims to eliminate errors commonly associated with paper-based processes, such as calculation mistakes and billing issues, while maintaining accurate records of all transactions.

The Bus Reservation System will include several key modules: seat availability inquiry, ticket reservation, ticket cancellation, login/authentication, and reservation history management. Each module is designed to streamline the operations of Jac Liner Inc., making it easier for ticket master to book and manage travel plans for passengers and for the company to handle reservations and maintain records.

This document is intended for use by the project team, stakeholders, and developers involved in the creation and implementation of the bus reservation system. It provides a clear and detailed description of the system's requirements, facilitating effective communication and ensuring that all parties have a mutual understanding of the project's goals and deliverables. The successful implementation of this system is expected to significantly enhance the efficiency and reliability of Jac Liner Inc.'s reservation processes, leading to improved customer satisfaction and operational effectiveness.

The system will be an on-site platform located at the bus terminal, with the bus selection applicable to a specific terminal, hence having a fixed departure terminal.

Introduction

Commuting is a common social activity for most Filipinos, serving as the primary mode of transportation between various destinations. However, it is also fraught with challenges. According to an editorial published in the Philippine Daily Inquirer titled "Commuters Deserve Better," public commuting in the Philippines is lamented as "dehumanizing." This sentiment underscores the pressing need for improvements in public transportation.

In parallel, there is a widely held belief in many international circles that we are currently living in the "information age," characterized by an explosion in the collection, processing, and sharing of information (Ajere et al., 2015). This shift has prompted managers and information consumers, especially those in the transportation sector, to seek a wider variety of information to enhance operations and management. Consequently, there is a growing demand for robust data and information management systems to meet these evolving needs.

The advent of digitalization has revolutionized many aspects of daily life, including how businesses operate and how public services are accessed. In the realm of public transportation, technologies play a crucial role in addressing these challenges. For instance, advanced reservation systems streamline booking processes, provide real-time information on schedules and seat availability, and enhance overall customer experience.

For ticket master working at bus terminals, an on-site reservation system offers significant benefits. It allows for efficient management of bookings and provides passengers with accurate information, reducing the stress and inconvenience of commuting. This system ensures that the bus selection is specific to a particular terminal, with a fixed departure point, simplifying the booking process and improving operational efficiency.

By leveraging technology to improve public transportation, ticket master can better meet the needs of commuters. This aligns with the broader digital transformation trend that seeks to enhance efficiency, convenience, and customer

satisfaction across various sectors. Advanced bus reservation systems are a crucial part of this transformation, offering Filipinos a more efficient way to travel.

Platform

Operating System: Windows
Technologies Use:
Front End: HTML, CSS, JavaScript, and Java Swing
Language: Java
Back End: MySQL
Hardware Requirements:
a. PC with Intel Celeron Processor (or above)
b. 4 GB RAM or above
c. 40 GB Hard Disk or above
Software Requirements:
a. Operating system : Windows
b. Front end : HTML, CSS, JavaScript, and Java Swing
c. Language: Java
d. Back end : MySQL

1. Introduction

1.1 Purpose

The purpose of the bus reservation system is to enhance the process of bus travel for the ticket master. By providing an on-site booking platform at the terminal, the system offers ticket master a convenient way to reserve seats, access schedules, and provide real-time updates to passengers without needing physical paperwork.

Furthermore, real-time tracking by the system improves accuracy by preventing overbooking and providing current information on bus schedules, seat availability, and travel conditions.

1.2 Scope

The Bus Reservation System covers the development and implementation of a computerized system for a bus terminal to manage data efficiently. The primary objective is to streamline all transactions, ensuring speed, accuracy, and elimination of errors. This system is designed to replace all manual paperwork, providing a reliable method for maintenance, ultimately ensuring the successful implementation of the computerized bus reservation system.

The Bus Reservation System consists of four main modules:

1. **User Authentication Module -** This module manages user authentication and authorization for the bus reservation system. Ticket master can log in using their credentials, ensuring that only authorized users can access the system's features. This maintains secure and restricted access to personal and booking information.
2. **Ticket Reservation Module** - This module facilitates the reservation of bus tickets. It allows ticket master to select the destination bus terminal, travel dates, and specific bus trips based on schedule and seat availability. The system provides details such as Bus IDs, origin and destination terminals, bus type (Regular or Deluxe), departure and arrival times, seat availability, and fares. The module updates the database with reservation details upon successful booking and includes a real-time seat availability inquiry to prevent overbooking and provide accurate information to users.
3. **Ticket Cancellation Module** - This module allows ticket master to cancel previously reserved tickets. It requires the ticket number for cancellation to track and delete the reservation accurately. Upon cancellation, the module updates the database by removing the relevant reservation details. This ensures accurate tracking and

management of cancellations, allowing the system to maintain up-to-date records and availability.

4. **Reservation History Management Module** - This module provides ticket master with a comprehensive view of their past reservation activities. It displays details of previous reservations, including ticket ID, bus ID, travel dates, destination terminals, departure and arrival times, and the number of seats reserved. This module helps the ticket master to keep track of reservation histories and manage records, enhancing overall operational efficiency.

1.3 Definitions, Acronyms, and Abbreviations

1. **Bus Reservation System (BRS)**: A software application that allows users to book, modify, and cancel bus tickets online.
2. **Bus ID:** A unique identifier assigned to each bus in the system.
3. **Ticket Master:** An individual responsible for handling the bus reservation system, managing ticket sales, ensuring proper seat allocation, and handling customer inquiries and complaints.
4. **Ticket Master ID:** A unique identifier assigned to a ticket master within the bus reservation system.
5. **Reservation**: The process of booking a bus ticket for a specific route and time.
6. **Bus Selection:** The process by which passengers choose a specific bus for their journey from a list of available buses.
7. **Seat Selection:** The process by which passengers choose specific seats on a selected bus.
8. **Seat Availability**: The number of seats available for booking on a particular bus route and schedule.
9. **Cancellation**: The process of voiding a previously made reservation.
10. **Discounted**: It refers to a bus ticket that is sold at a reduced price compared to the regular fare.
11. **Schedule**: The timetable of bus departures and arrivals.

12. **Route**: The path or course taken by a bus from its starting point to its destination.
13. **Route ID**: A unique identifier assigned to a specific bus route within the bus reservation system.
14. **Ticket**: A document or electronic record that confirms a seat reservation on a bus.
15. **Ticket ID:** A unique identifier assigned to each ticket issued by the system.

Technology Overview

The Bus Reservation System (BRS) utilizes various essential technologies to ensure its operation is flexible, efficient, and user-friendly:

1. **Java:** It is used for back-end development to write core application logic, handle business processes, and manage server-side operations.
2. **XAMPP:** It is a development environment that bundles Apache, MySQL, etc., for local development and testing purposes.
    - **Apache HTTP Server:** It is a reliable, secure, and flexible web server that serves web pages, handles requests, manages static content, and integrates with Java for serving dynamic content.
    - **MySQL:** It is a relational database management system used for storing and retrieving data related to users, reservations, schedules, routes, and more, featuring SQL querying, transactions, and data integrity.
3. **HTML/CSS/JavaScript/Java Swing**: It is used for creating the front-end user interface, ensuring a responsive and interactive experience.
4. **Bootstrap**: It is for responsive and mobile-first front-end web development.
5. **Git**: It is for version control, enabling collaborative development and tracking of code changes.

2. *Over-all Project Description*

2.1 *Project/Product Perspective*

This on-site Bus Reservation System streamlines bookings and manages schedules and seat availability for ticket master.

2.2 *Project/product Functions*

The system functionalities include:

- **User Authentication:**
  Ticket master will log into the system using their credentials to access the booking and management features, ensuring that only authorized personnel can use the system.
- **Trip Selection:**
  Ticket master will use the reservation page to input the passenger's choice of destination terminal from the drop-down menu. They will also enter the date of the bus trip.
- **Bus Selection:**
  After completing the terminal and date selection, the system generates a series of bus trips from which the ticket master will choose according to the passenger's choice. This includes details such as Bus IDs, destination terminals, types (AC or non-AC), scheduled departure and arrival times, seat availability, and corresponding fares.
- **Seat Selection:**
  Based on the number of available seats on the chosen bus, ticket master will select the seat(s) for the passenger from the available options.
- **Seat Indicator:**
  Ticket master will select seats for passengers, with a functionality to indicate seats reserved for persons with disabilities (PWDs) and others who require special assistance. Passengers requiring these seats may present identification to confirm their eligibility.

- **Ticket Cancellation:**
  After a successful reservation, a ticket is issued to the passenger. If the passenger wishes to cancel their reservation, ticket master will use the ticket number to track and delete the reservation in the system.
- **Reservation History Management:**
  The system maintains a history of all reservations made by the ticket master. This includes details such as ticket ID, bus ID, date of travel, destination bus terminals, departure and arrival times, and the number of seats reserved for each trip. This allows ticket master to review and manage past reservations efficiently.

2.3 User Classes and Characteristics

The system will be used in a Bus Terminal, with the Ticket Master being the primary user. The system is designed to be user-friendly and incorporates a Graphical User Interface (GUI).

- **Ticket Master:** Individuals responsible for managing and overseeing the bus reservation system. Ticket master use the system to handle bookings, cancellations, and seat allocations, as well as to assist passengers with inquiries and resolve issues. They require a moderate level of computer literacy and familiarity with the system's functionalities to perform their duties efficiently.

2.4 Operating Environment

The bus reservation system operates within a specified environment that includes essential hardware and software components. Hardware requirements encompass servers for system hosting, client devices such as PCs and mobile devices for user interaction, and peripherals like printers for ticket issuance. Software needs include a server operating system such as Windows Server or Linux, compatibility with common web browsers like

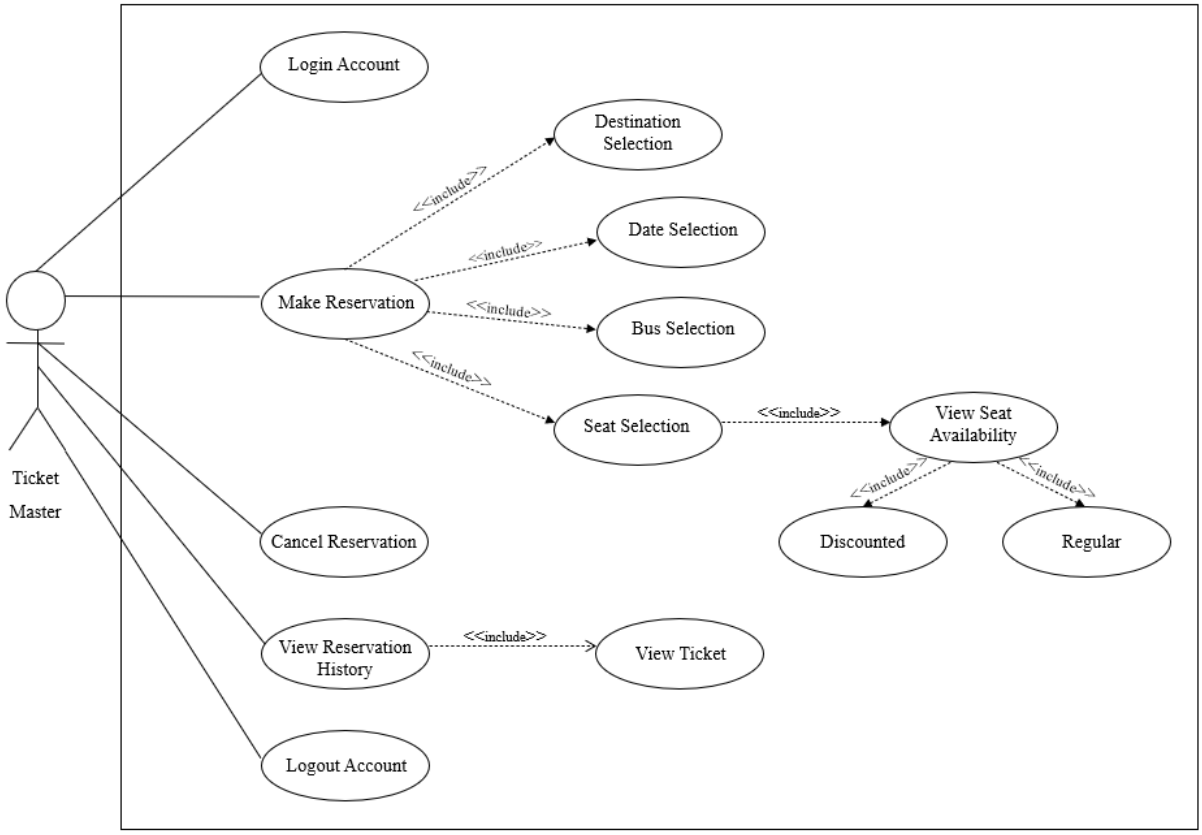
Chrome and Firefox for client access, and database management systems such as MySQL or PostgreSQL for efficient data storage and management. Integration with payment gateways and adherence to data protection regulations are critical considerations. This environment aims to deliver a user-friendly interface for both customers and administrators, ensuring reliable performance under varying operational loads.

2.5 Constraints

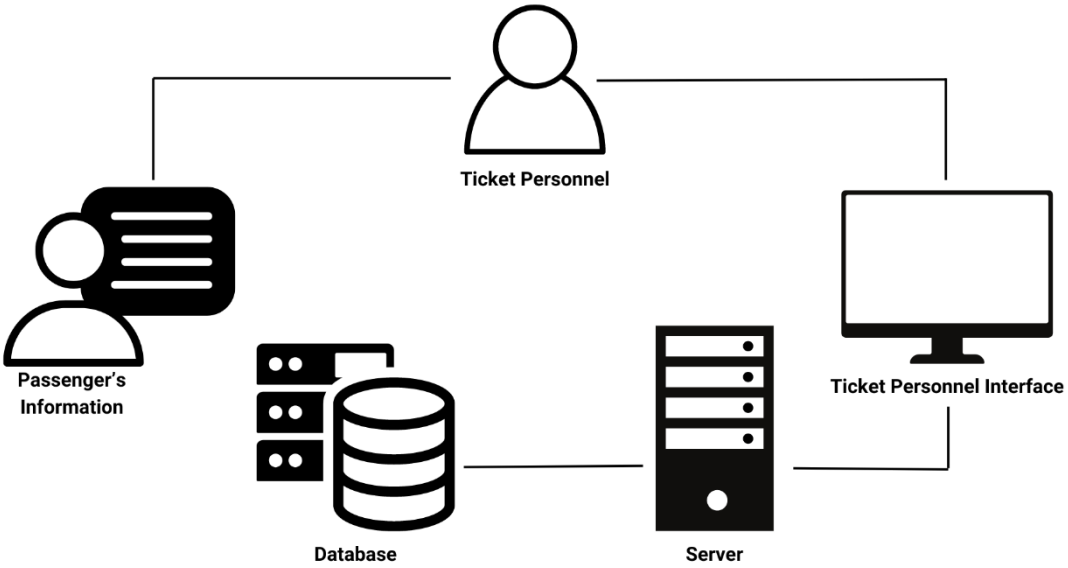
In this study, the researchers developed a bus reservation system which the ticket master uses to manage available bus routes, allocate seats, and book tickets on-site for passengers. However, due to operational constraints, the system is subject to the following limitations:

- The system facilitates seat reservations and ticket printing. It does not currently support online payments through this platform.
- This system operates with a fixed small screen size and does not support full-screen mode.
- The system lacked mechanisms for users to provide feedback and seek support, such as service contacts, help documentation, and user forums.

2.6 Use case diagram/model

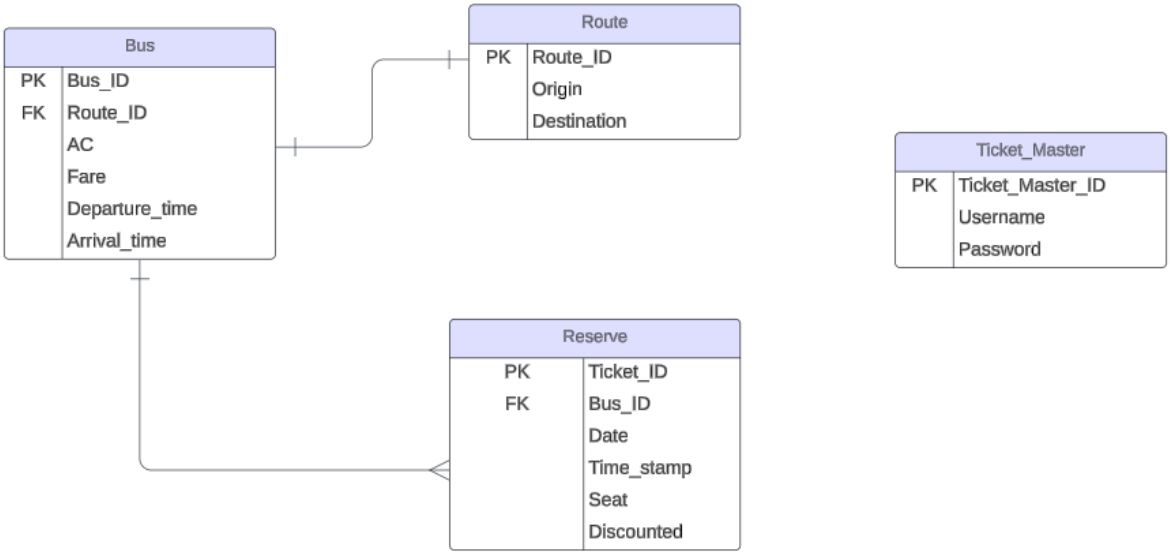


2.7 System Architecture



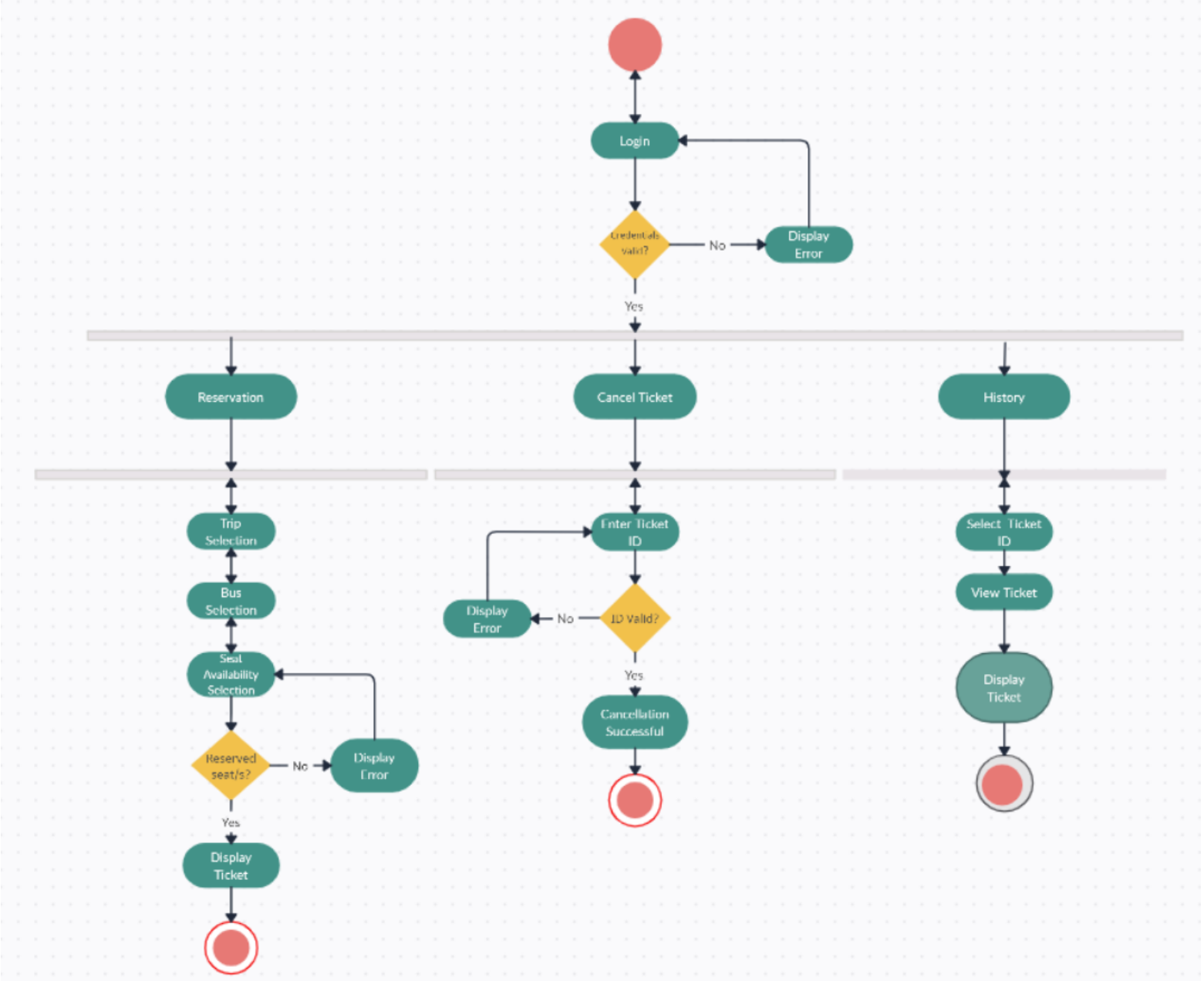
3. Data Design

3.1 Entity Relationship Diagram



4. Functional Design Description

4.1 Activity Diagram



5. Conclusion

The Bus Reservation System designed for Jac Liner Inc. represents a significant advancement in the management of bus reservations and related operations. By transitioning from manual, paper-based processes to a computerized system, this project aims to enhance the speed, accuracy, and efficiency of the company's reservation transactions.

The system includes essential modules such as ticket reservation, ticket cancellation, and travel history management. These modules work together to streamline operations, reduce errors, and improve overall customer satisfaction. The system ensures that all transactions are accurately recorded, thereby eliminating common issues associated with manual processes, such as calculation mistakes and billing inaccuracies.

Implementing this system is expected to provide numerous benefits to both Jac Liner Inc. and its customers. For the company, it offers a more efficient way to manage reservations, maintain accurate records, and improve operational effectiveness.

In conclusion, the Bus Reservation System is a vital tool for Jac Liner Inc. to modernize its operations, enhance customer service, and achieve greater efficiency. The successful deployment of this system will mark a significant step forward in the company's ongoing efforts to improve its services and remain competitive in the transportation industry.

6. References

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